

Solve financial and real-world modelling problems

Solve problems involving money, budgets, best buys, measurement, time, and other real-world situations.

Step 57

★ Today I am learning to solve financial and real-world modelling problems.

1. Money and Budgeting

Solve each problem.

- a) Amira has £42.75. She spends £18.60 on books and £7.45 on a lunch. She then saves the rest. How much money does she save?

Answer: £ _____

- b) A family has a weekly budget of £350 for food, transport and entertainment. They spend £142.80 on food and £68.65 on transport. How much is left for entertainment?

Answer: £ _____

- c) A jacket costs £67.50. It is on sale with 20% off. What is the sale price?

Answer: £ _____

- d) Mia buys a pair of jeans for £28.00 and a top for £17.50. She pays with a £50 note. How much change does she get?

Answer: £ _____

Think about it!

- Read carefully and identify what you need to find.
- Show your working.
- Include the correct units (£, kg, L, km, hours, etc.).
- Check your answer makes sense.

2. Best Buys

Which is the better value for money? Show your working.

- a) Pack A: 500 g of cereal for £1.65
 Pack B: 750 g of cereal for £2.25
 Which pack is better value?

Answer: _____

- c) Packet M: 250 mL of juice for 85p
 Packet N: 400 mL of juice for £1.20
 Which packet is better value?

Answer: _____

- b) Store X: 12 rolls of toilet paper for £5.40
 Store Y: 18 rolls of toilet paper for £7.20
 Which store is better value?

Answer: _____

- d) Bag P: 1.5 kg of apples for £2.10
 Bag Q: 2 kg of apples for £2.70
 Which bag is better value?

Answer: _____

Think about it!

- Find the unit price (price per 1 unit).
- Compare the unit prices.
- Choose the option with the lowest unit price.

3. Time and Work

Solve each problem.

- a) A train leaves London at 09:45 and arrives in Manchester at 13:30. How long is the journey?

 Answer: _____
 (hours and minutes)

- b) A machine can make 360 widgets in 6 hours. How many widgets can it make in 1 hour? In 8 hours?

 Answer: _____ per hour
 _____ in 8 hours

- c) 8 workers can paint a fence in 9 days. If take 6 workers to paint the same fence? work at the same rate.

Answer: _____ days

Think about it!

- Use appropriate formulas:
 $\text{time} = \frac{\text{distance}}{\text{speed}}$
 $\text{rate} = \frac{\text{work}}{\text{time}}$
 $\text{work} = \text{rate} \times \text{time}$
- Keep your units consistent.
- Round answers where needed.

4. Measurement and Conversion

Solve each problem.

- a) A recipe needs 175 L of milk. You only have a measuring jug in millilitres. How many millilitres of milk do you need?

Answer: _____ mL

- b) A room is 5.4 m long and 3.8 m wide. A carpet costs £18.50 per m². How much will it cost to buy enough carpet to fit the room?

Answer: £ _____

- c) A box has a mass of 2.6 kg. It contains 8 apples. What is the average mass of one apple?

Answer: _____ kg

- d) A tank holds 1,800 L of water. It is filled to 65% of its capacity. How many litres of water are in the tank?

Answer: _____ L

Think about it!

- 1 L = 1,000 mL
- Area = length × width
- Average = $\frac{\text{total}}{\text{number of items}}$
- Percentage of an amount = $\left(\frac{p}{100}\right) \times \text{amount}$

5. Percentages and Discounts

Solve each problem.

- a) A television is reduced from £420 to £336. What is the percentage decrease?

Answer: _____ %

- b) A shop increases the price of a game from £28 to £33.60. What is the percentage increase?

Answer: _____ %

- c) A laptop costs £750. A store offers 15% off and then an extra 5% off the sale price. What is the final price?

Answer: £ _____

- d) A jacket is on sale for £64 after a 20% discount. What was the original price?

Think about it!

- Increase: $\text{New} = \text{Original} \times \left(1 + \frac{p}{100}\right)$
- Decrease: $\text{New} = \text{Original} \times \left(1 - \frac{p}{100}\right)$
- For successive discounts, apply them one after the other.
- Use a calculator to check.

6. Multi-step Real-World Problems

Solve each problem.

- a) A family goes on a road trip. They drive 420 km on the first day and 380 km on the second day. They have 250 km left to reach their destination. What is the total distance of the trip?

Answer: _____ km

- b) A café makes £1250 in sales on Saturday and £980 on Sunday. It spends £420 on ingredients and £190 on other costs. How much profit did it make over the weekend?

Answer: £ _____

- c) Sara invests £2,000 in an account that earns 4.5% interest per year. How much interest will she earn in 3 years? (Assume interest is simple.)

Answer: £ _____

- d) A school bus can carry 28 students. There are 96 students going on a trip. Each student pays £12 for the trip. How many buses are needed and how much money is collected in total?

 Answer: _____ buses
 £ _____ in total

Think about it!

- Break the problem into steps.
- Do one step at a time.
- Check that your answer answers the question.

7. Open-Ended Modelling

Create your own model and solve.

- a) Design a budget for a class event with a total of £200. Include at least three items with different costs. Show how much money is left over.

Answer: _____

- b) Plan a trip where you need to travel a distance using two different types of transport (e.g. bus and train). Find the total time and cost.

Answer: _____

- c) You want to save £500 in 6 months. Create a saving plan with the same amount saved each month. How much will you save each month?

Answer: _____

- d) Create your own real-world problem involving money, time or measurement. Solve it and explain your answer.

Answer: _____

Think about it!

- Be realistic.
- Use numbers that make sense.
- Show all your working.
- Explain your answer clearly.

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Understands concepts
☐ Uses number skills accurately
☐ Solves problems confidently

- ☐ Explains thinking clearly
☐ Needs more practice

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____